

Lab 10 - Evaluation : Text Analysis with NLP

In this evaluation, we perform a complete NLP analysis pipeline on **Amazon product reviews**.

Dataset: `amazon_review.csv`

Tasks:

1. Sentiment analysis + bar chart visualization
2. Text preprocessing (lowercase, punctuation, stopwords, lemmatization)
3. Top 15 most frequent words (bar chart)
4. Word cloud of lemmatized reviews
5. POS frequency: nouns, verbs, adjectives

Dataset loaded: 50 reviews, 2 columns
Columns: ['reviewText', 'overall']

	reviewText	overall
0	Absolutely love this product! Works perfectly ...	5
1	Great quality for the price. Very happy with m...	5
2	This is the best buy I have made in a long tim...	5
3	Exceeded my expectations. The build quality is...	5
4	Amazing product! Does exactly what it promises...	5
5	Fantastic value. My kids love it and it is ver...	5
6	Top notch quality. Easy to set up and works fl...	5
7	I am so happy with this purchase! Perfect for ...	5
8	Excellent product. Delivery was fast and packa...	5
9	Works great! Very satisfied with the performan...	5

Question 1 : Sentiment Analysis + Bar Chart

Perform a sentiment analysis on the product reviews, classify each one as `positive`, `negative`, or `neutral`, and generate a bar chart to visualize the distribution of these sentiments.

Question 2 : Text Preprocessing

Convert the text to lowercase, remove punctuation and numbers, filter out common stopwords, and lemmatize the remaining words using their appropriate POS tags.

Preprocessing pipeline example:

Original : Absolutely love this product! Works perfectly and arrived fast.
Processed: ['absolutely', 'love', 'product', 'work', 'perfectly', 'arrive', 'fast']

Original : Great quality for the price. Very happy with my purchase.
Processed: ['great', 'quality', 'price', 'happy', 'purchase']

Original : This is the best buy I have made in a long time. Highly recommended!
Processed: ['best', 'buy', 'make', 'long', 'time', 'highly', 'recommend']

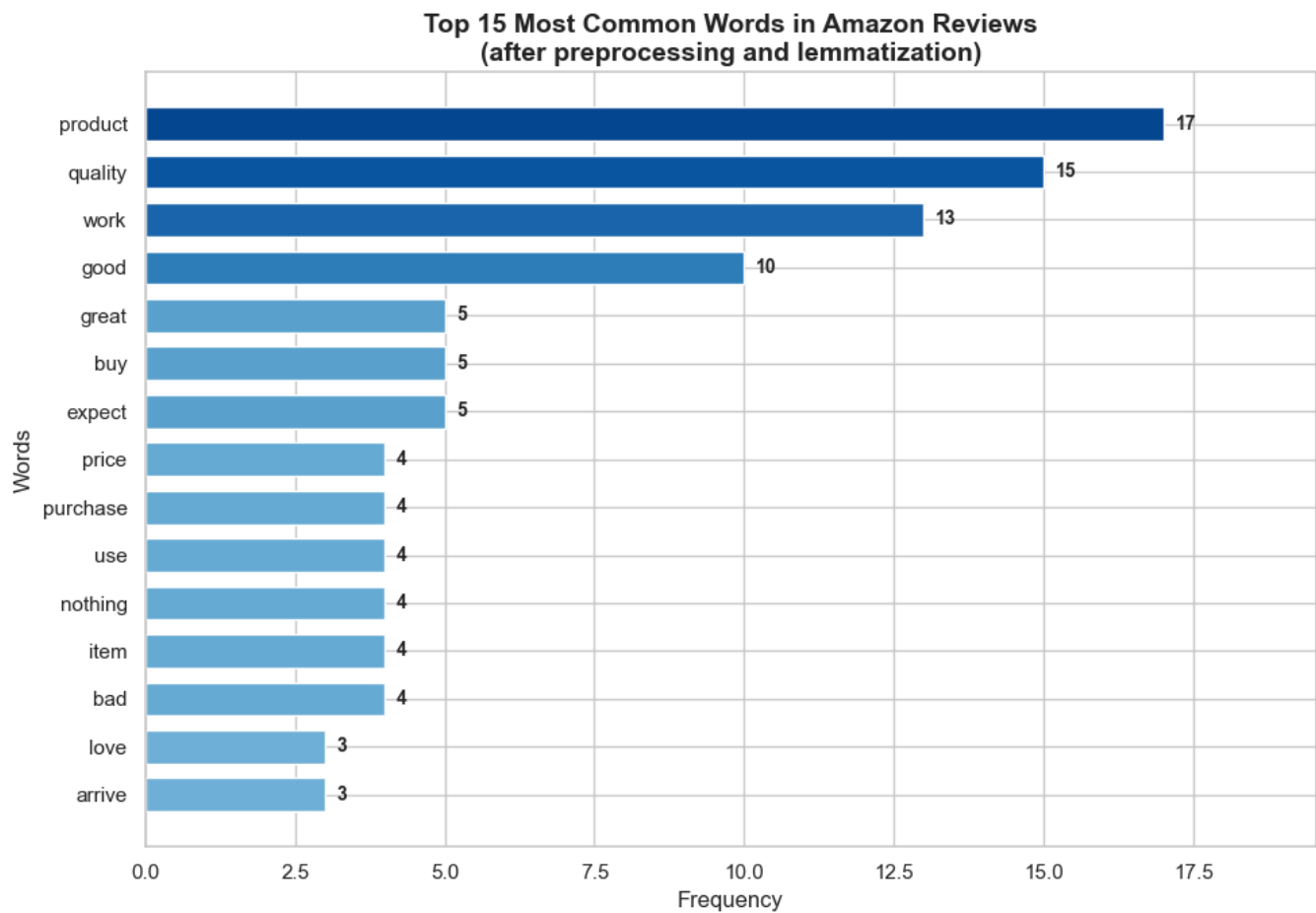
Total tokens after preprocessing : 290
Unique tokens : 157

Question 3 : Top 15 Most Common Words (Bar Chart)

After preprocessing and lemmatization, calculate the frequency of each unique word and display a bar chart showing the **top 15 most common words**.

Top 15 most frequent words after preprocessing:

1. product	:	17	#####
2. quality	:	15	#####
3. work	:	13	#####
4. good	:	10	#####
5. great	:	5	#####
6. buy	:	5	#####
7. expect	:	5	#####
8. price	:	4	####
9. purchase	:	4	####
10. use	:	4	####
11. nothing	:	4	####
12. item	:	4	####
13. bad	:	4	####
14. love	:	3	###
15. arrive	:	3	###



Question 4 : Word Cloud

Generate a word cloud based on the entire corpus of lemmatized review words to provide a visual summary of the most prominent terms.

A word cloud visualization of customer feedback for a product. The words are arranged in a circular pattern, with 'quality' and 'product' being the most prominent. Other significant words include 'work', 'price', 'great', 'expect', 'nothing', 'job', 'advertising', 'absolutely', 'acceptable', 'happy', 'perfectly', 'purchase', 'excellent', 'small', 'long', 'useful', 'secure', 'mixed', 'mediocre', 'happy', 'break', 'sturdy', 'time', 'look', 'bad', 'easy', 'highly', 'solid', 'special', 'others', 'perfect', 'feel', 'nice', 'money', 'average', 'love', 'immediately', 'exactly', 'outstanding', 'recommend', 'arrive', 'fast', 'clear', 'poor', 'item', 'fine', 'serious', 'amazing', 'make', 'well', 'overall', 'condition', 'promise', 'different', 'decent', 'cheap', 'satisfied', 'expectation', 'quite', 'minor', 'still', 'arrive', 'recommen', 'impressive', 'service', 'good', 'flawless', 'pretty', 'feeling', 'excellent', 'small', 'long', 'useful', 'secure', 'mixed', 'mediocre', 'happy', 'break', 'sturdy', 'time', 'look', 'bad', 'easy', 'highly', 'solid', 'special', 'others', 'perfect', 'feel', 'nice', 'money', 'average', 'love', 'immediately', 'exactly', 'outstanding', 'recommend', 'arrive', 'fast', 'clear', 'poor', 'item', 'fine', 'serious', 'amazing', 'make', 'well', 'overall', 'condition', 'promise', 'different', 'decent', 'cheap', 'satisfied', 'expectation', 'quite', 'minor', 'still', 'arrive', 'recommen'.

Question 5 : POS Frequency — Nouns, Verbs, Adjectives

Nouns : 119 tokens, 58 unique
Verbs : 60 tokens, 34 unique
Adjectives : 84 tokens, 53 unique

The figure consists of three horizontal bar charts, each representing a different part of speech. The x-axis for all charts is 'Frequency'.

- Top 12 Nouns (119 total):** The y-axis lists nouns, and the x-axis shows frequency from 0 to 16. The bars are orange.

Noun	Frequency
product	17
quality	15
price	4
purchase	4
nothing	4
item	4
use	3
work	3
performance	3
job	3
money	3
time	2
- Top 12 Verbs (60 total):** The y-axis lists verbs, and the x-axis shows frequency from 0 to 10. The bars are blue.

Verb	Frequency
work	10
expect	5
buy	4
love	3
arrive	3
make	2
recommend	2
look	2
break	2
advertise	2
stop	2
exceed	1
- Top 12 Adjectives (84 total):** The y-axis lists adjectives, and the x-axis shows frequency from 0 to 8. The bars are green.

Adjective	Frequency
good	9
great	5
bad	4
happy	3
best	2
long	2
durable	2
easy	2
excellent	2
fine	2
nice	2
small	2

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good      : 9x (positive, polarity=0.70)
great     : 5x (positive, polarity=0.80)
bad       : 4x (negative, polarity=-0.70)
happy     : 3x (positive, polarity=0.80)
best      : 2x (positive, polarity=1.00)
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